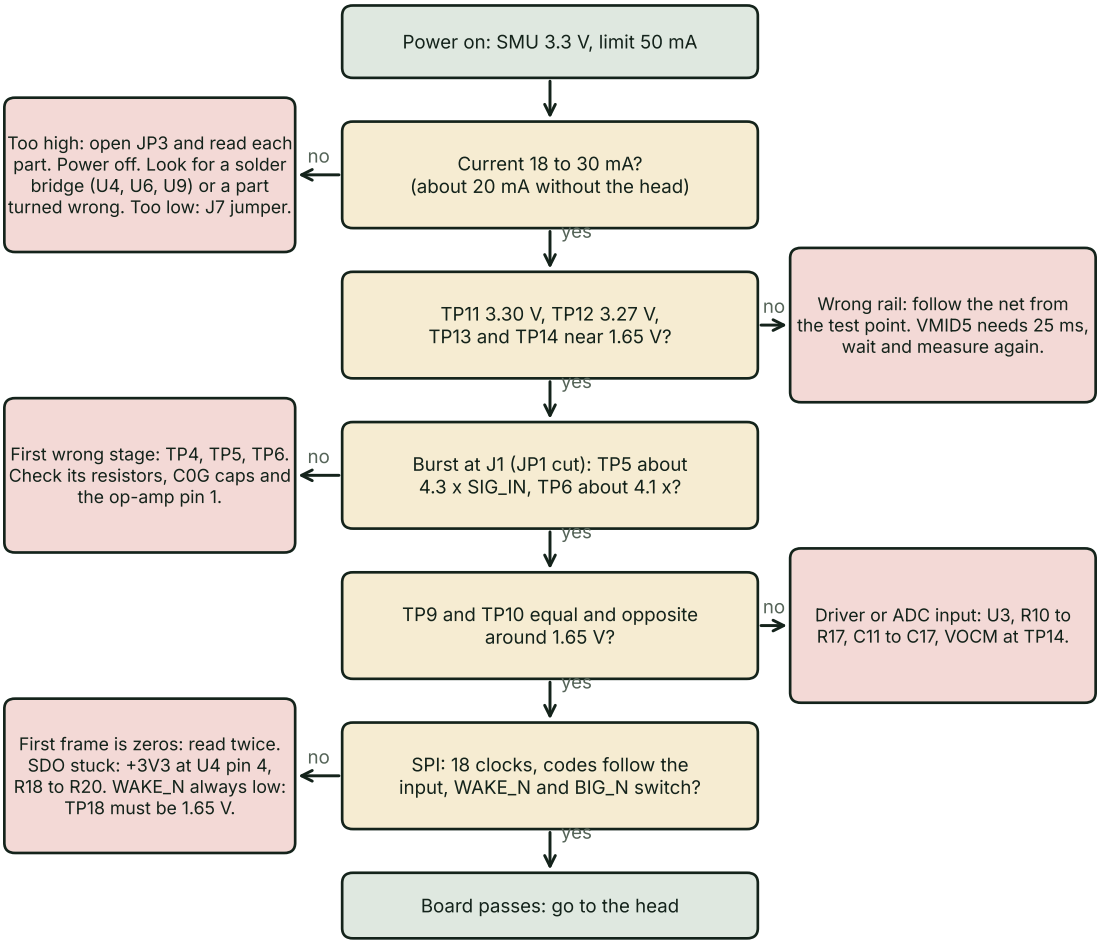
Power on: SMU 3.3 V, limit 50 mA
Current 18 to 30 mA?
(about 20 mA without the head)
no
Too high: open JP3 and read each part. Power off. Look for a solder bridge (U4, U6, U9) or a part turned wrong. Too low: J7 jumper.
yes
TP11 3.30 V, TP12 3.27 V,
TP13 and TP14 near 1.65 V?
no
Wrong rail: follow the net from the test point. VMID5 needs 25 ms, wait and measure again.
yes
Burst at J1 (JP1 cut): TP5 about
4.3 x SIG_IN, TP6 about 4.1 x?
no
First wrong stage: TP4, TP5, TP6. Check its resistors, C0G caps and the op-amp pin 1.
yes
TP9 and TP10 equal and opposite
around 1.65 V?
no
Driver or ADC input: U3, R10 to R17, C11 to C17, VOCM at TP14.
yes
SPI: 18 clocks, codes follow the
input, WAKE_N and BIG_N switch?
no
First frame is zeros: read twice. SDO stuck: +3V3 at U4 pin 4, R18 to R20. WAKE_N always low: TP18 must be 1.65 V.
yes
Board passes: go to the head